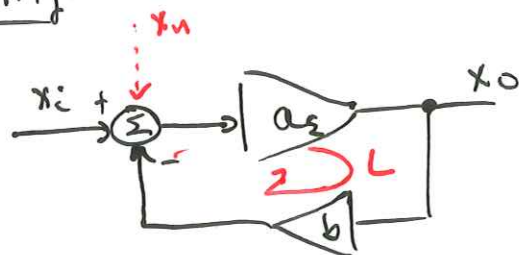


Stability



a_z is positive & real

b is positive & real

such that $L = a_z b$ is positive & real

$$x_o = a_z \left(x_i - \frac{x_o}{b} \right)$$

discrete consideration

let $x_i(k) = 0$,

this is not fully correct!

if $x_o(k=1) \neq 0$, then $x_o(k=2) = -L x_o(k=1)$
but really more like... $x_o(k=2) = x_o(k=1) - L x_o(k=1)$

what actually happens: because of the - sign,

$x_o(k \dots)$ will decay.

such that $x_o(k \rightarrow \infty) \rightarrow 0$ if $x_i(k) = 0$

Good!
stable

what if a_z OR b is negative?

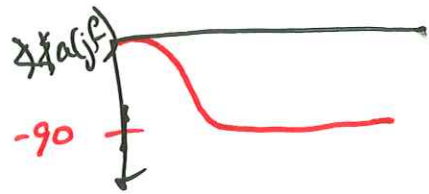
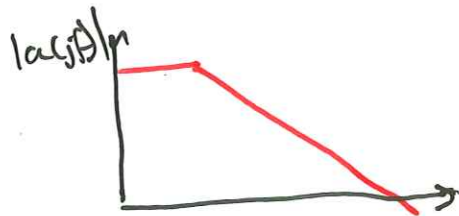
$\rightarrow L$ is now negative: effective positive feedback!

$\rightarrow x_o(k \rightarrow \infty) \rightarrow \pm \infty$ if $x_i(k) = 0$

Bad!
unstable

Dominant Pole Amp

simple model

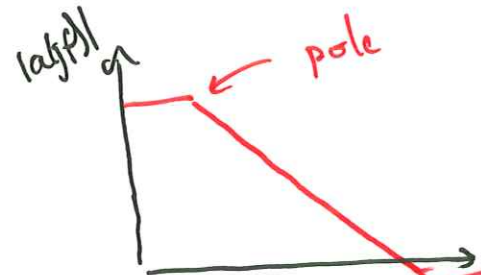


$a(jf)$ will never be negative!
(but can be fully imaginary)

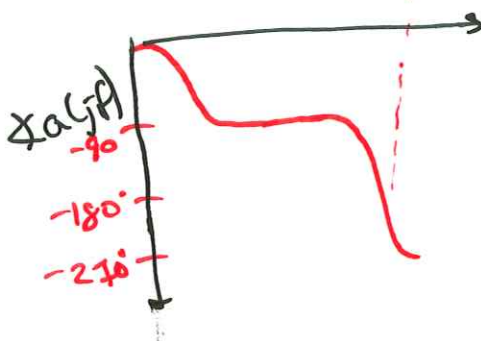
if β is positive $\rightarrow$ stable!

$T(jf) \rightarrow$ never negative $\hat{z}$, real stable!

less simple model:



additional poles



$a(jf)$ at higher frequencies can have $\angle a(jf) = -180^\circ$

$\rightarrow a(jf)$ can be real $\hat{z}$ negative!
 $\rightarrow @ f_{-180a}$

if β is positive

at $f_{-180a} \rightarrow T(jf_{-180a})$ is real

$\hat{z}$ negative
 $\rightarrow$ possibly unstable!

(at this frequency)

Dominant pole added such that $|a(jf_{-180a})|$ very small!
or at least $< 0.5\beta$

Consider effects of $T(j\omega)$ on $A(j\omega)$

$$A(j\omega) = A_0 / (1 + 1/T(j\omega)) = A_0 [1 / (1 + 1/T(j\omega))]$$

if $T(j\omega)$ is real & positive,

$$1 / (1 + 1/T(j\omega)) \leq 1 \quad |A(j\omega)| \leq |A_0|$$

if $T(j\omega)$ is real & negative

$1 / (1 + 1/T(j\omega))$... can be in 3 regimes.

1) $T(j\omega) = -1$

$$1 / (1 + 1/-1) = 1 / (1 - 1) = 1/0 = \pm\infty \rightarrow |A(j\omega)| = \infty \leftarrow \text{not necessarily true}$$

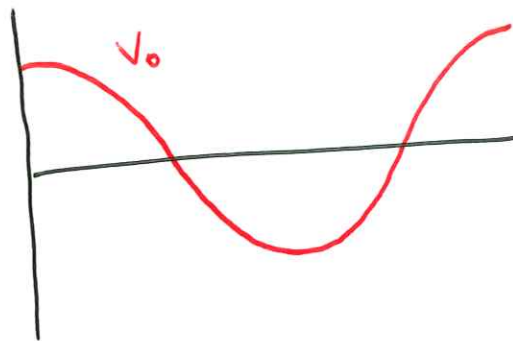
if $V_i = 0$, then V_o is undefined.

$$(V_o = V_i A(j\omega) = \frac{0}{0})$$

if $V_i \neq 0 \rightarrow V_o \rightarrow \infty$

* single impulse of noise in V_i

Any other amount of input
would drive it to infinity.



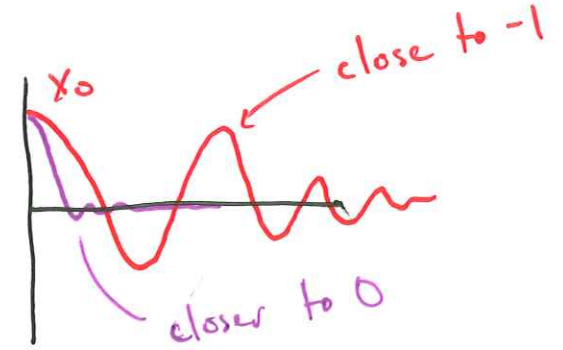
2)

~~$0 \geq T(jf) \geq -1$~~

$0 \geq T(jf) \geq -1$

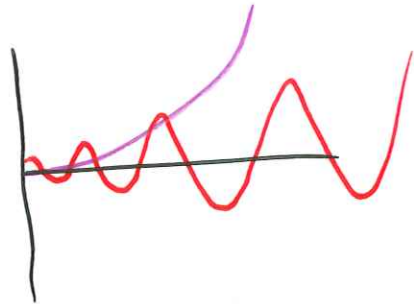
loop gain is small, any persistent signal decays around the loop

$T(jf)$ is real & negative, but small acceptable!



3) $T(jf) < -1$ ($|T(jf)| > 1$)

→ any existing signal will grow uncontrollably



Since $T(jf)$ will be real and negative at some ~~frequency~~ frequency (ies) then we need to ensure regime (2) above.

Characterizing Stability

point at which $\angle T(jf) = -180^\circ \rightarrow f_{-180}$

at this frequency, calculate $|T(jf_{-180})|$

→ "Gain Margin" $GM = -20 \log_{10} |T(jf_{-180})|$

if $GM > 0\text{dB} \rightarrow \text{stable}$

$GM \leq 0\text{dB} \rightarrow \text{unstable}$

point at which $|T(jf)| = 1 \rightarrow f_x$

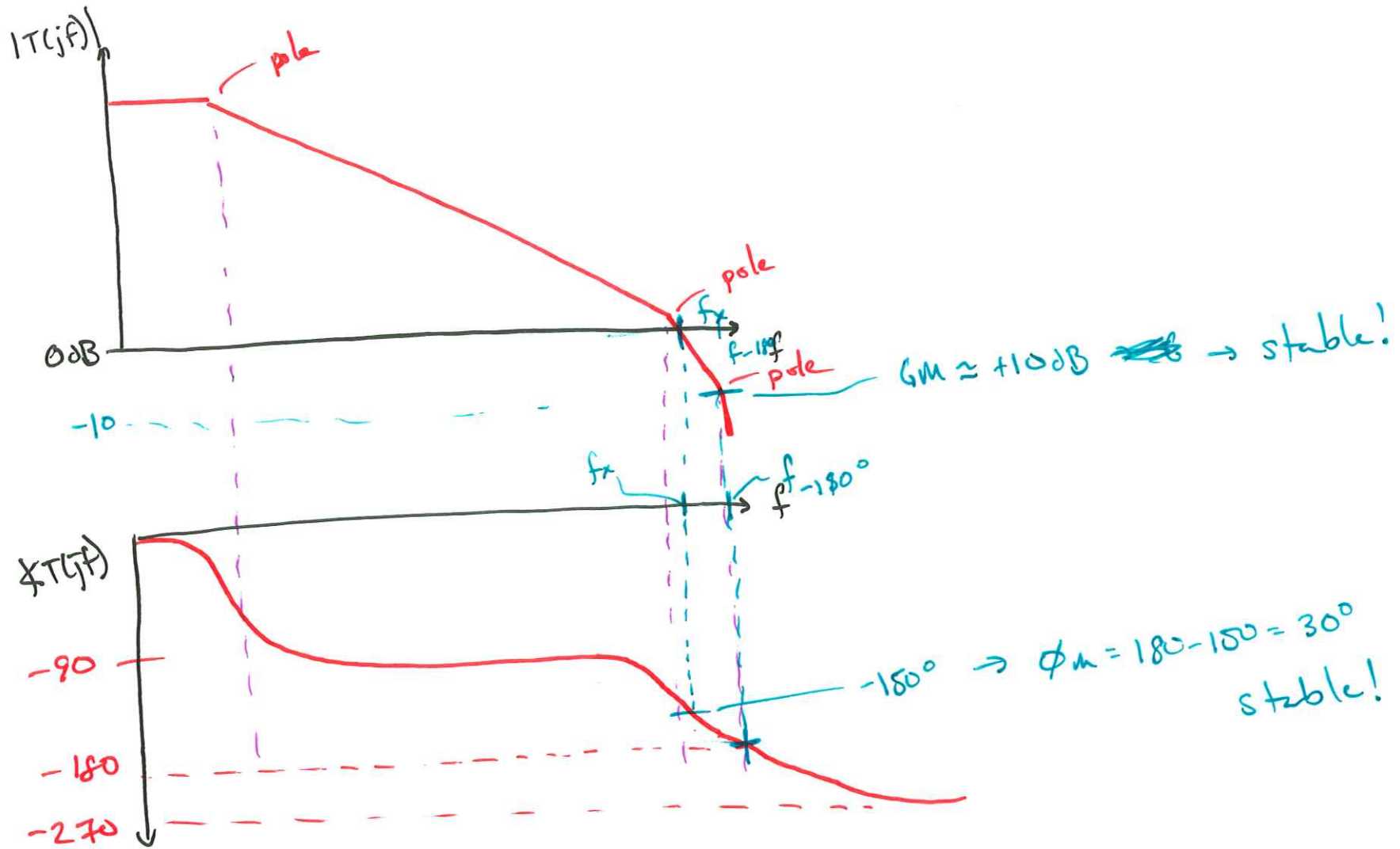
at this frequency, calculate $\angle T(jf_x)$

→ "Phase Margin" $\phi_m = 180^\circ + \angle T(jf_x)$

$\phi_m > 0^\circ \rightarrow \text{stable}$

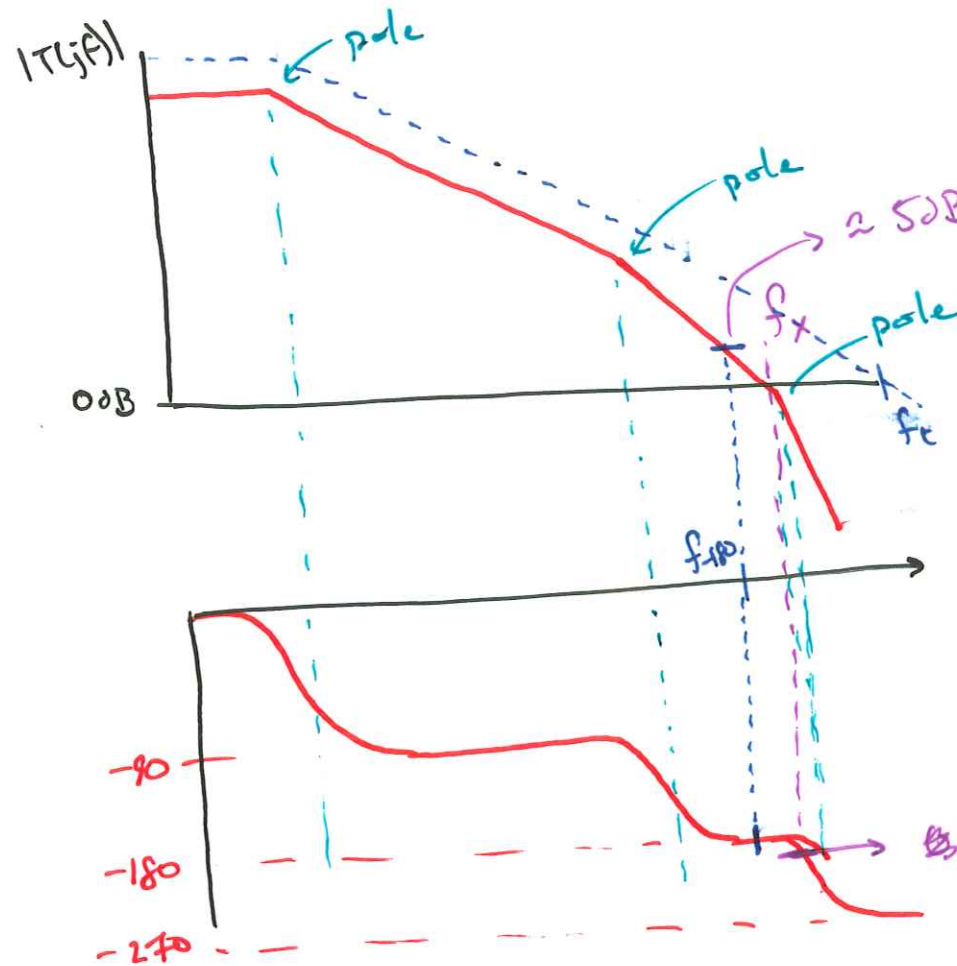
$\phi_m \leq 0^\circ \rightarrow \text{unstable.}$

Example 1 of G_M, ϕ_M



Example 2 of GM , ϕ_m

$$A(j\omega) = A_0 / (1 + j\omega T(j\omega))$$



$\approx 50B \rightarrow GM = -50B \rightarrow \text{unstable!}$

$\phi \approx -190 \rightarrow \phi_m = 180^\circ - 190^\circ = -10^\circ$
unstable!

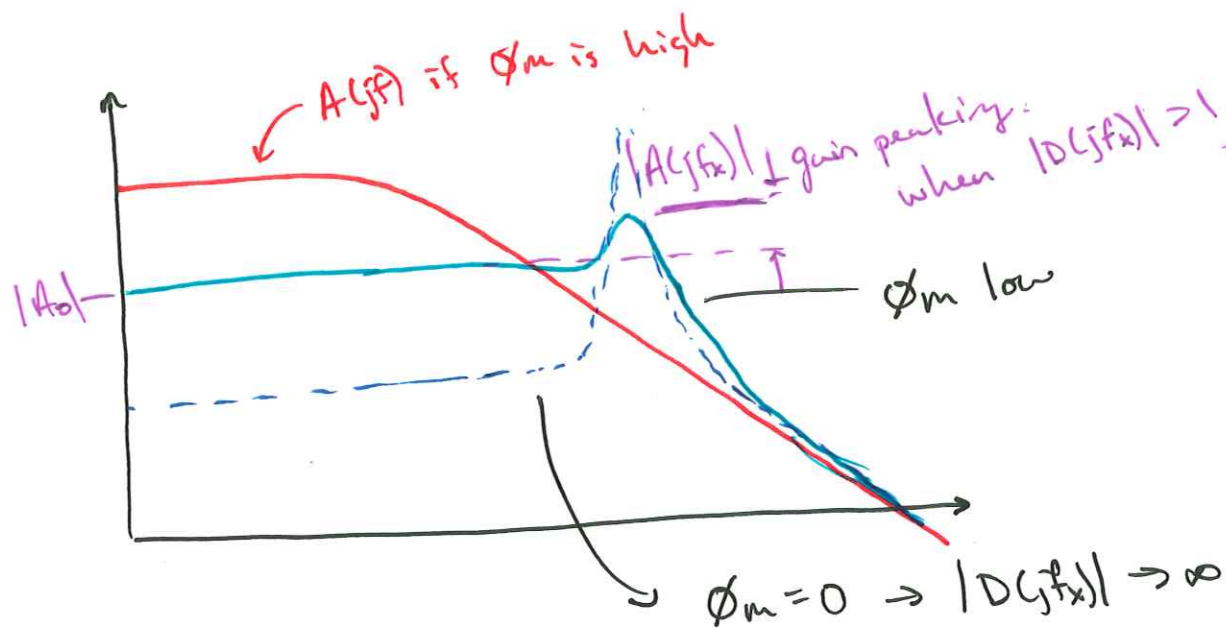
$$A(j\omega) \approx \frac{A_0}{1 + 1/T(j\omega)}$$

$$D(j\omega) = \frac{1}{1 + 1/T(j\omega)} \rightarrow \text{Discrepancy factor}$$

$$A(j\omega) \approx A_0 D(j\omega) \quad \text{then at } f_x, D(jf_x) = \frac{1}{1 + 1/e^{j\phi_m}} = \frac{1}{1 + 1/e^{j\phi_m}}$$

$D(jf_x) \rightarrow$ measure gain error @ f_x

$$|D(jf_x)| = \frac{1}{\sqrt{2(1 - \cos\phi_m)}}$$



$$|A(jf_x)| = |A_0| |D(jf_x)|$$

$$\frac{|A(jf_x)|}{|A_0|} = |D(jf_x)|$$